Homework 3

Andrew Hwang
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Haohua Deng

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1 3.1

Question 15. Find the solution of the initial value problem

$$2y'' - 3y' + y = 0, \qquad y(0) = 2, \quad y'(0) = \frac{1}{2}.$$

Then determine the maximum value of the solution and also find the point where the solution is zero.

Solution. We are given the characteristic equation

$$2r^2 - 3y + 1 = 0.$$

Hence,

$$r = \frac{3 \pm \sqrt{9 - 4(2)}}{4} = 1, \frac{1}{2}.$$

Hence, $y_1 = e^t$ and $y_2 = e^{t/2}$ are solutions, and we are given the general solution

$$y = Ay_1 + By_2 = Ae^t + Be^{t/2}, \quad A, B \in \mathbb{R}.$$

Now let us solve the initial values, we are given a system of equation

$$\begin{cases} y(t) = Ae^t + Be^{t/2} \\ y'(t) = Ae^t + \frac{B}{2}e^{t/2} \end{cases}$$

At $t = 0$, we get the system

$$\begin{cases} y(0) = A + B = 2 \\ y'(0) = A + \frac{B}{2} = \frac{1}{2} \implies 2A + B = 1 \end{cases}$$

Subtracting the first equation from the second equation gives $A = -1$, and substituting this back onto the first equation gives $B = 3$. Hence, the solution to the IVP is

$$\boxed{y = -e^t + 3e^{t/2}.}$$

To find the maximum, we set $y'(t) = 0$:

$$y'(t) = -e^t + \frac{3}{2}e^{t/2} = 0$$

$$e^{t/2}\left(-e^{t/2} + \frac{3}{2}\right) = 0$$

Since $e^{t/2} > 0$ for all t, we need

$$-e^{t/2} + \frac{3}{2} = 0 \implies e^{t/2} = \frac{3}{2} \implies t = 2\ln\left(\frac{3}{2}\right)$$

Substituting back into $y(t)$:

$$y\left(2\ln\frac{3}{2}\right) = -e^{2\ln(3/2)} + 3e^{\ln(3/2)} = -\left(\frac{3}{2}\right)^2 + 3\left(\frac{3}{2}\right) = -\frac{9}{4} + \frac{9}{2} = \boxed{\frac{9}{4}}.$$

Lastly, to find the zeros, we set $y(t) = 0$:

$$-e^t + 3e^{t/2} = 0$$

$$e^{t/2}\left(-e^{t/2} + 3\right) = 0.$$

Since $e^{t/2} > 0$, we look for

$$-e^{t/2} + 3 = 0 \implies e^{t/2} = 3 \implies t = 2\ln|3|.$$

Therefore the point is:

$$\boxed{(2\ln|3|, 0)}$$

$\square$

2 3.2

Question 8. Determine the longest interval in which the given initial value problem is certain to have a unique twice-differentiable solution. Do not attempt to find the solution.

$$y'' + (\cos t)\,y' + 3(\ln|t|)\,y = 0, \qquad y(2) = 3, \quad y'(2) = 1$$

Solution. To find the uniqueness of the solution, we must check that the functions $p(t) = \cos t$, $q(t) = 3\ln|t|$, and $g(t) = 0$ are continuous. It is obvious that $p(t)$ and $g(t)$ are continuous everywhere, but $q(t)$ does not exist at $t = 0$. Hence, because the initial value is at $t = 2$, based on the theorem, the longest interval is $\boxed{(0, \infty).}$ $\square$

Question 22. Consider the equation $y'' - y' - 2y = 0$.

(a) Show that $y_1(t) = e^{-t}$ and $y_2(t) = e^{2t}$ form a fundamental set of solutions.

(b) Let $y_3(t) = -2e^{2t}$, $y_4(t) = y_1(t) + 2y_2(t)$, and $y_5(t) = 2y_1(t) - 2y_3(t)$. Are $y_3(t)$, $y_4(t)$, and $y_5(t)$ also solutions of the given differential equation?

(c) Determine whether each of the following pairs forms a fundamental set of solutions:

$$\{y_1(t),\, y_3(t)\}; \quad \{y_2(t),\, y_3(t)\}; \quad \{y_1(t),\, y_4(t)\}; \quad \{y_4(t),\, y_5(t)\}.$$

Solution. For part (a), the characteristic equation is

$$r^2 - r - 2 = 0,$$

and hence

$$r = \frac{1 \pm \sqrt{1 + 4(1)(2)}}{2} = 2, -1.$$

Therefore, $y_1(t) = e^{-t}$ and $y_2(t) = e^{2t}$ are solutions. To confirm they form a fundamental set, we compute the Wronskian:

$$W(y_1, y_2)(t) = \begin{vmatrix} e^{-t} & e^{2t} \\ -e^{-t} & 2e^{2t} \end{vmatrix} = 2e^t + e^t = 3e^t \neq 0$$

Since $W \neq 0$ for all t, y_1 and y_2 form a fundamental set of solutions. For part (b), we recognize that $y_3(t) = -2e^{2t} = -2y_2$. Hence, $y_3(t)$ is a solution because it can be represented as a linear combination of y_1 and y_2. Similarly, this holds for y_4. Now, since y_5 is a linear combination of y_1 and y_3, we can write:

$$y_5(t) = 2y_1(t) - 2y_3(t) = 2y_1(t) - 2(-2y_2(t)) = 2y_1(t) + 4y_2(t).$$

Since y_5 can also be expressed as a linear combination of y_1 and y_2, it is also a solution. Therefore y_3, y_4, and y_5 are all solutions. For part (c), we take a linear algebra approach. A pair forms a basis if and only if the two solutions are linearly independent, i.e., neither is a scalar multiple of the other. We check each pair:

- $\{y_1(t),\, y_3(t)\}$: $y_3 = -2y_2$, not a multiple of y_1, so **linearly independent.** ✓
- $\{y_2(t),\, y_3(t)\}$: $y_3 = -2y_2$, so they are scalar multiples of each other, so **linearly dependent.**
- $\{y_1(t),\, y_4(t)\}$: $y_4 = y_1 + 2y_2$, not a multiple of y_1, so **linearly independent.**✓
- $\{y_4(t),\, y_5(t)\}$: $y_4 = y_1 + 2y_2$ and $y_5 = 2y_1 + 4y_2 = 2(y_1 + 2y_2) = 2y_4$, so they are scalar multiples of each other, so **linearly dependent.**

Hence, $\{y_1(t),\, y_3(t)\}$ and $\{y_1(t),\, y_4(t)\}$ form a basis (i.e. a fundamental set). □

Question 25. Find the Wronskian of two solutions of the given differential equation without solving the equation.

$$(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0, \qquad \text{(Legendre's equation)}$$

Solution. Using Abel's Theorem, the Wronskian is given by

$$W[y_1, y_2] = c\exp\left(-\int p(x)\,dx\right).$$

Dividing the equation by $(1 - x^2)$ to get standard form gives $p(x) = \dfrac{-2x}{1 - x^2}$, so

$$W[y_1, y_2] = c\exp\left(-\int \frac{-2x}{1 - x^2}\,dx\right).$$

Let $u = 1 - x^2$, then $du = -2x\,dx$. Hence:

$$\int \frac{-2x}{1 - x^2}\,dx = \int \frac{du}{u} = \ln|u| = \ln|1 - x^2|.$$

Therefore:

$$W[y_1, y_2] = c\exp\left(-\ln|1 - x^2|\right) = \boxed{\frac{c}{1 - x^2}}$$

□

3 3.3

Question 5. Find the general solution of the given differential equation.

$$y'' - 2y' + 2y = 0$$

Solution. The characteristic equation is $r^2 - 2r + 2 = 0$, so

$$r = \frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i.$$

Hence we have complex roots with real part $\lambda = 1$ and imaginary part $\mu = 1$. Using Euler's formula, the two real-valued linearly independent solutions are

$$y_1 = e^t \cos t, \qquad y_2 = e^t \sin t.$$

We can verify these are linearly independent by computing the Wronskian:

$$W(y_1, y_2) = \begin{vmatrix} e^t \cos t & e^t \sin t \\ e^t(\cos t - \sin t) & e^t(\sin t + \cos t) \end{vmatrix} = e^{2t}(\sin t + \cos t)\cos t - e^{2t}(\cos t - \sin t)\sin t = e^{2t} \neq 0.$$

Since $W \neq 0$, the general solution is:

$$\boxed{y = e^t(c_1 \cos t + c_2 \sin t), \qquad c_1, c_2 \in \mathbb{R}.}$$

$\square$

Question 18. Consider the initial value problem

$$y'' + 2y' + 6y = 0, \qquad y(0) = 2, \quad y'(0) = \alpha \geq 0.$$

(a) Find the solution $y(t)$ of this problem.

(b) Find α such that $y = 0$ when $t = 1$.

(c) Find, as a function of α, the smallest positive value of t for which $y = 0$.

(d) Determine the limit of the expression found in part (c) as $\alpha \to \infty$.

Solution. For part (a), the characteristic equation is $r^2 + 2r + 6 = 0$, so

$$r = \frac{-2 \pm \sqrt{4 - 24}}{2} = \frac{-2 \pm \sqrt{-20}}{2} = -1 \pm i\sqrt{5}.$$

Hence we have complex roots with $\lambda = -1$ and $\mu = \sqrt{5}$, giving the general solution:

$$y = e^{-t}(c_1 \cos(\sqrt{5}\, t) + c_2 \sin(\sqrt{5}\, t)).$$

Applying $y(0) = 2$ gives

$$c_1 = 2.$$

Differentiating gives

$$y' = -e^{-t}(c_1 \cos(\sqrt{5}\, t) + c_2 \sin(\sqrt{5}\, t)) + e^{-t}(-c_1\sqrt{5} \sin(\sqrt{5}\, t) + c_2\sqrt{5} \cos(\sqrt{5}\, t)),$$

so applying $y'(0) = \alpha$ results in

$$-c_1 + c_2\sqrt{5} = \alpha \implies c_2 = \frac{\alpha + 2}{\sqrt{5}}.$$

Therefore the solution is:

$$\boxed{y = e^{-t}\left(2\cos(\sqrt{5}\, t) + \frac{\alpha + 2}{\sqrt{5}} \sin(\sqrt{5}\, t)\right).}$$

For part (b), we set $y(1) = 0$:

$$e^{-1}\left(2\cos(\sqrt{5}) + \frac{\alpha + 2}{\sqrt{5}} \sin(\sqrt{5})\right) = 0.$$

Since $e^{-1} \neq 0$:

$$2\cos(\sqrt{5}) + \frac{\alpha + 2}{\sqrt{5}} \sin(\sqrt{5}) = 0 \implies \alpha + 2 = -\frac{2\sqrt{5}\cos(\sqrt{5})}{\sin(\sqrt{5})}$$

$$\alpha = -2\sqrt{5}\cot(\sqrt{5}) - 2 \approx \boxed{1.51.}$$

For part (c), we set $y(t) = 0$. Since $e^{-t} \neq 0$:

$$2\cos(\sqrt{5}\,t) + \frac{\alpha+2}{\sqrt{5}}\sin(\sqrt{5}\,t) = 0$$

$$\tan(\sqrt{5}\,t) = -\frac{2\sqrt{5}}{\alpha+2}$$

$$\sqrt{5}\,t = \pi - \arctan\left(\frac{2\sqrt{5}}{\alpha+2}\right)$$

$$\boxed{t = \frac{1}{\sqrt{5}}\left(\pi - \arctan\left(\frac{2\sqrt{5}}{\alpha+2}\right)\right).}$$

For part (d), as $\alpha \to \infty$, $\dfrac{2\sqrt{5}}{\alpha+2} \to 0$, so $\arctan\left(\dfrac{2\sqrt{5}}{\alpha+2}\right) \to 0$. Therefore:

$$\lim_{\alpha\to\infty} t = \frac{1}{\sqrt{5}}(\pi - 0) = \boxed{\frac{\pi}{\sqrt{5}}.}$$

$\square$

4 3.4

Question 12. Consider the following modification of the initial value problem in Example 2:

$$y'' - y' + \frac{y}{4} = 0, \qquad y(0) = 2, \quad y'(0) = b.$$

Find the solution as a function of b, and then determine the critical value of b that separates solutions that remain positive for all $t > 0$ from those that eventually become negative.

Solution. The characteristic equation is

$$r^2 - r + \frac{1}{4} = 0 \implies \left(r - \frac{1}{2}\right)^2 = 0 \implies r = \frac{1}{2} \text{ (repeated root).}$$

Hence the general solution is

$$y = (c_1 + c_2 t)e^{t/2}.$$

Applying $y(0) = 2$ gives

$$c_1 = 2.$$

Differentiating the general solution gives

$$y' = c_2 e^{t/2} + \frac{1}{2}(c_1 + c_2 t)e^{t/2},$$

and by applying $y'(0) = b$,

$$c_2 + \frac{c_1}{2} = b \implies c_2 + 1 = b \implies c_2 = b - 1.$$

Therefore the solution as a function of b is

$$\boxed{y = (2 + (b-1)t)\, e^{t/2}.}$$

Now, we find the critical value. Since $e^{t/2} > 0$ for all t, the sign of y is determined entirely by the factor $2 + (b-1)t$. By algebra, we get that $\boxed{b = 1}$ is the critical value. This can be further proven by checking limiting cases outside $b = 1$.

1. If $b > 1$, then $c_2 = b - 1 > 0$, so $2 + (b-1)t \to \infty$ as $t \to \infty$ — the solution remains positive for all $t > 0$.

2. If $b < 1$, then $c_2 = b - 1 < 0$, so $2 + (b-1)t$ is a decreasing linear function which eventually becomes negative at

$$t' = \frac{2}{1-b} > 0.$$

□

Question 15.

(a) Consider the equation $y'' + 2ay' + a^2y = 0$. Show that the roots of the characteristic equation are $r_1 = r_2 = -a$ so that one solution of the equation is e^{-at}.

(b) Use Abel's formula [equation (23) of Section 3.2] to show that the Wronskian of any two solutions of the given equation is

$$W(t) = y_1(t)y_2'(t) - y_1'(t)y_2(t) = c_1e^{-2at},$$

where c_1 is a constant.

(c) Let $y_1(t) = e^{-at}$ and use the result of part (b) to obtain a differential equation satisfied by a second solution $y_2(t)$. By solving this equation, show that $y_2(t) = te^{-at}$.

Proof. For part (a), the characteristic equation is

$$r^2 + 2ar + a^2 = 0 \implies (r+a)^2 = 0 \implies r_1 = r_2 = -a.$$

Since we have a repeated root $r = -a$, one solution is $\boxed{y_1(t) = e^{-at}.}$ □

For part (b), in standard form given $p(t) = 2a$, the Wronskian is

$$W(t) = c_1 \exp\left(-\int p(t)\,dt\right) = c_1 \exp\left(-\int 2a\,dt\right) = c_1\exp(-2at) = \boxed{c_1e^{-2at}.}$$

□

For part (c), let $y_1(t) = e^{-at}$. The Wronskian condition from part (b) gives

$$W(t) = y_1y_2' - y_1'y_2 = c_1e^{-2at}.$$

Substituting $y_1 = e^{-at}$ and $y_1' = -ae^{-at}$ gives

$$e^{-at}y_2' - (-ae^{-at})y_2 = c_1e^{-2at}$$

$$e^{-at}y_2' + ae^{-at}y_2 = c_1e^{-2at}.$$

Dividing through by e^{-at} yields

$$y_2' + ay_2 = c_1e^{-at},$$

a first-order linear ODE. The integrating factor is $\mu = e^{at}$, so

$$\int \frac{d}{dt}\left[e^{at}y_2\right] = \int c_1$$

$$e^{at}y_2 = c_1t + c_2 \implies y_2 = c_1te^{-at} + c_2e^{-at}.$$

Since $y_1 = e^{-at}$ is already y_1, the second linearly independent solution is

$$\boxed{y_2(t) = te^{-at}.}$$

□

Question 21. Use the method of reduction of order to find a second solution of the given differential equation.

$$xy'' - y' + 4x^3y = 0, \qquad x > 0, \qquad y_1(x) = \sin(x^2)$$

Solution. Let $y_2 = v(x)y_1(x) = v(x)\sin(x^2)$. Then

$$y_2' = v'\sin(x^2) + 2xv\cos(x^2),$$

$$\begin{aligned} y_2'' &= v''\sin(x^2) + 2xv'\cos(x^2) + 2xv'\cos(x^2) + 2v\cos(x^2) - 4x^2v\sin(x^2) \\ &= v''\sin(x^2) + 4xv'\cos(x^2) + 2v\cos(x^2) - 4x^2v\sin(x^2). \end{aligned}$$

Substituting into original differential equation gives

$$x\left[v''\sin(x^2) + 4xv'\cos(x^2) + 2v\cos(x^2) - 4x^2v\sin(x^2)\right]$$

$$-\left[v'\sin(x^2) + 2xv\cos(x^2)\right] + 4x^3v\sin(x^2) = 0$$

$$xv''\sin(x^2) + 4x^2v'\cos(x^2) + \cancel{2xv\cos(x^2)} - \cancel{4x^3v\sin(x^2)} - v'\sin(x^2) - \cancel{2xv\cos(x^2)} + \cancel{4x^3v\sin(x^2)} = 0$$

$$xv''\sin(x^2) + v'\left(4x^2\cos(x^2) - \sin(x^2)\right) = 0.$$

Let $w = v'$, so $w' = v''$:

$$x\sin(x^2)w' + \left[4x^2\cos(x^2) - \sin(x^2)\right]w = 0.$$

Now, we have reduced the equation to a first order linear equation. We now find $w = v'$:

$$\int \frac{dw}{w} = \int \frac{\sin(x^2) - 4x^2\cos(x^2)}{x\sin(x^2)}\,dx = \int \left(\frac{1}{x} - \frac{4x\cos(x^2)}{\sin(x^2)}\right)dx.$$

$$\ln|w| = \ln|x| - 2\ln|\sin(x^2)| = \ln\left|\frac{x}{\sin^2(x^2)}\right|.$$

Therefore,

$$w = v' = \frac{x}{\sin^2(x^2)}.$$

Integrating to find v, let $u = x^2$, $du = 2x\,dx$, so

$$v = \int \frac{x}{\sin^2(x^2)}\,dx = \frac{1}{2}\int \frac{du}{\sin^2 u} = \frac{1}{2}\int \csc^2 u\,du = -\frac{1}{2}\cot(x^2).$$

Therefore, the second solution is

$$y_2 = v(x)y_1(x) = -\frac{1}{2}\cot(x^2)\sin(x^2) = -\frac{1}{2}\cos(x^2) \implies \boxed{y_2 = \cos(x^2).}$$

□

Question 28. If a, b, and c are positive constants, show that all solutions of

$$ay'' + by' + cy = 0$$

approach zero as $t \to \infty$.

Proof. The characteristic equation is

$$ar^2 + br + c = 0,$$

with roots:

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

We consider three cases based on the discriminant $\Delta = b^2 - 4ac$.
Case 1: $\Delta > 0$ (two distinct real roots). Both roots are real, so

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

Since $a, b, c > 0$, we have $\sqrt{b^2 - 4ac} < \sqrt{b^2} = b$, so both roots satisfy $r_1, r_2 < 0$. The general solution is:

$$y = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

Since $r_1, r_2 < 0$, both terms decay to zero, so $y \to 0$ as $t \to \infty$.
Case 2: $\Delta = 0$ (repeated real root). The repeated root is

$$r = -\frac{b}{2a} < 0$$

since $a, b > 0$. The general solution is

$$y = (c_1 + c_2 t)e^{-bt/2a}.$$

Since the exponential decay dominates the linear factor as $t \to \infty$,

$$\lim_{t \to \infty} (c_1 + c_2 t)e^{-bt/2a} = 0.$$

Case 3: $\Delta < 0$ (complex conjugate roots). The roots are

$$r = -\frac{b}{2a} \pm i\frac{\sqrt{4ac - b^2}}{2a} = \lambda \pm i\mu,$$

where $\lambda = -\dfrac{b}{2a} < 0$ since $a, b > 0$. The general solution is

$$y = e^{\lambda t}(c_1 \cos(\mu t) + c_2 \sin(\mu t)).$$

Since $\lambda < 0$, the factor $e^{\lambda t} \to 0$ as $t \to \infty$, and the trigonometric terms are bounded, so $y \to 0$ as $t \to \infty$. In all three cases, every solution approaches zero as $t \to \infty$. $\square$

5 3.5

Question 10. Find the general solution of the given differential equation.

$$y'' + y' + 4y = 2\sinh t$$

Solution. First, the homogeneous equation $y'' + y' + 4y = 0$ has characteristic equation

$$r^2 + r + 4 = 0 \implies r = \frac{-1 \pm \sqrt{1-16}}{2} = \frac{-1 \pm i\sqrt{15}}{2}.$$

Hence the homogeneous solution is

$$y_h = e^{-t/2}\left[c_1 \cos\left(\frac{\sqrt{15}}{2}t\right) + c_2 \sin\left(\frac{\sqrt{15}}{2}t\right)\right].$$

Next, we guess a particular solution of the form:

$$Y(t) = Ae^t + Be^{-t}.$$

Then,

$$Y(t)' = Ae^t - Be^{-t}, \qquad Y(t)'' = Ae^t + Be^{-t}.$$

Substituting into the equation gives

$$(Ae^t + Be^{-t}) + (Ae^t - Be^{-t}) + 4(Ae^t + Be^{-t}) = e^t - e^{-t}.$$

$$6Ae^t + 4Be^{-t} = e^t - e^{-t}.$$

By matching the terms, we get

$$6A = 1 \implies A = \frac{1}{6}, \qquad 4B = -1 \implies B = -\frac{1}{4}.$$

Therefore, the particular solution is

$$Y(t) = \frac{1}{6}e^t - \frac{1}{4}e^{-t}.$$

Now, we have the homogeneous general solution and the particular solution. Therefore, the general solution is

$$\boxed{y = e^{-t/2}\left[c_1 \cos\left(\frac{\sqrt{15}}{2}t\right) + c_2 \sin\left(\frac{\sqrt{15}}{2}t\right)\right] + \frac{1}{6}e^t - \frac{1}{4}e^{-t}.}$$

□

Question 13. Find the solution of the given initial value problem.

$$y'' - 2y' + y = te^t + 4, \qquad y(0) = 1, \quad y'(0) = 1$$

Solution. First, the homogeneous equation $y'' - 2y' + y = 0$ has characteristic equation

$$r^2 - 2r + 1 = 0 \implies (r-1)^2 = 0 \implies r = 1 \text{ (repeated root).}$$

Hence the homogeneous solution is

$$y_h = (c_1 + c_2 t)e^t.$$

Next, we split the right side into two parts and find a particular solution for each.
Part 1: $y'' - 2y' + y = 4$. We guess $g_1 = A$, so $g_1' = g_1'' = 0$. Then,

$$0 - 0 + A = 4 \implies A = 4.$$

Therefore $g_1 = 4$.
Part 2: $y'' - 2y' + y = te^t$. Since $r = 1$ is a repeated root, e^t and te^t are both homogeneous solutions. We must multiply by t^2 and guess

$$g_2 = t^2(At + B)e^t = (At^3 + Bt^2)e^t.$$

Then,

$$\begin{aligned} g_2' &= (3At^2 + 2Bt)e^t + (At^3 + Bt^2)e^t = (At^3 + Bt^2 + 3At^2 + 2Bt)e^t \\ g_2'' &= (At^3 + Bt^2 + 3At^2 + 2Bt)e^t + (3At^2 + 2Bt + 6At + 2B)e^t \\ &= (At^3 + Bt^2 + 6At^2 + 4Bt + 6At + 2B)e^t. \end{aligned}$$

Substituting into $y'' - 2y' + y = te^t$ and canceling e^t gives

$$(\cancel{At^3} + \cancel{Bt^2} + \cancel{6At^2} + \cancel{4Bt} + 6At + 2B) - 2(\cancel{At^3} + \cancel{Bt^2} + \cancel{3At^2} + \cancel{2Bt}) + (\cancel{At^3} + \cancel{Bt^2}) = t.$$

$$6At + 2B = t.$$

By matching coefficients,

$$6A = 1 \implies A = \frac{1}{6}, \qquad 2B = 0 \implies B = 0.$$

Therefore $g_2 = \frac{1}{6}t^3e^t$. The general solution is

$$y = (c_1 + c_2 t)e^t + \frac{1}{6}t^3e^t + 4.$$

Now we apply the initial conditions. From $y(0) = 1$,

$$c_1 + 4 = 1 \implies c_1 = -3.$$

$$y' = c_2e^t + (c_1 + c_2t)e^t + \frac{1}{2}t^2e^t + \frac{1}{6}t^3e^t.$$

From $y'(0) = 1$,

$$c_2 + c_1 = 1 \implies c_2 - 3 = 1 \implies c_2 = 4.$$

Therefore the particular solution is

$$\boxed{y = (-3 + 4t)e^t + \frac{1}{6}t^3e^t + 4.}$$

□

Question 15. Find the solution of the given initial value problem.

$$y'' + 2y' + 5y = 4e^{-t}\cos(2t), \qquad y(0) = 1, \quad y'(0) = 0$$

Solution. First, the homogeneous equation $y'' + 2y' + 5y = 0$ has characteristic equation

$$r^2 + 2r + 5 = 0 \implies r = \frac{-2 \pm \sqrt{4-20}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i.$$

Hence the homogeneous solution is

$$y_h = e^{-t}(c_1\cos(2t) + c_2\sin(2t)).$$

Next, for the particular solution we note that the right side is $4e^{-t}\cos(2t)$. Since $r = -1+2i$ *is* a root of the characteristic equation, we must multiply by t and guess

$$Y(t) = te^{-t}(A\cos(2t) + B\sin(2t)).$$

The derivatives ar

$$Y(t)' = e^{-t}\left[(A - At + 2Bt)\cos(2t) + (B - Bt - 2At)\sin(2t)\right],$$
$$Y(t)'' = e^{-t}\left[(-4At + 4B - 2A)\cos(2t) + (-4Bt - 4A - 2B)\sin(2t)\right].$$

Substituting into $y'' + 2y' + 5y = 4e^{-t}\cos(2t)$, and canceling e^{-t} gives

$$(-4At + 4B - 2A) + 2(A - At + 2Bt) + 5At)\cos(2t)$$
$$+((-4Bt - 4A - 2B) + 2(B - Bt - 2At) + 5Bt)\sin(2t) = 4\cos(2t).$$
$$4B\cos(2t) - 4A\sin(2t) = 4\cos(2t).$$

By matching coefficients,

$$4B = 4 \implies B = 1, \qquad -4A = 0 \implies A = 0.$$

Therefore the particular solution is

$$Y(t) = te^{-t}\sin(2t),$$

and the general solution is

$$y = e^{-t}(c_1\cos(2t) + c_2\sin(2t)) + te^{-t}\sin(2t).$$

Lastly, we solve boundary conditions. If $y(0) = 1$, $c_1 = 1$. Now,

$$y' = -e^{-t}(c_1\cos(2t)+c_2\sin(2t))+e^{-t}(-2c_1\sin(2t)+2c_2\cos(2t))+e^{-t}\sin(2t)+2te^{-t}\cos(2t)-te^{-t}\sin(2t).$$

Applying $y'(0) = 0$:

$$-c_1 + 2c_2 + 0 = 0 \implies -1 + 2c_2 = 0 \implies c_2 = \frac{1}{2}.$$

Therefore the solution is

$$\boxed{y = e^{-t}\left(\cos(2t) + \frac{1}{2}\sin(2t)\right) + te^{-t}\sin(2t).}$$

□

Question 23. Determine the general solution of

$$y'' + \lambda^2 y = \sum_{m=1}^{N} a_m \sin(m\pi t),$$

where $\lambda > 0$ and $\lambda \neq m\pi$ for $m = 1, \ldots, N$.

Solution. The homogeneous equation $y'' + \lambda^2 y = 0$ has characteristic equation

$$r^2 + \lambda^2 = 0 \implies r = \pm \lambda i.$$

Hence the homogeneous solution is

$$y_h = c_1 \cos(\lambda t) + c_2 \sin(\lambda t).$$

The summation may be deceiving, but since we can split solutions by superposition, it suffices to find a particular solution for each term $g_m(t) = a_m \sin(m\pi t)$ separately. For each $m = 1, \ldots, N$, consider

$$y'' + \lambda^2 y = a_m \sin(m\pi t).$$

Since $\lambda \neq m\pi$, the function $\sin(m\pi t)$ is *not* a homogeneous solution. We guess

$$Y_m(t) = A_m \sin(m\pi t).$$

Now, we compute the derivatives.

$$Y_m(t)' = A_m m\pi \cos(m\pi t), \qquad Y_m(t)'' = -A_m m^2\pi^2 \sin(m\pi t).$$

Substituting back into the original equation gives

$$-A_m m^2\pi^2 \sin(m\pi t) + \lambda^2 A_m \sin(m\pi t) = a_m \sin(m\pi t)$$

$$A_m(\lambda^2 - m^2\pi^2) = a_m \implies A_m = \frac{a_m}{\lambda^2 - m^2\pi^2}.$$

Therefore the particular solution for each term is

$$Y_m(t) = \frac{a_m}{\lambda^2 - m^2\pi^2} \sin(m\pi t).$$

By superposition, the total particular solution is

$$Y(t) = \sum_{m=1}^{N} Y_m(t) = \sum_{m=1}^{N} \frac{a_m}{\lambda^2 - m^2\pi^2} \sin(m\pi t),$$

and the general solution is

$$\boxed{y = c_1 \cos(\lambda t) + c_2 \sin(\lambda t) + \sum_{m=1}^{N} \frac{a_m}{\lambda^2 - m^2\pi^2} \sin(m\pi t).}$$

□